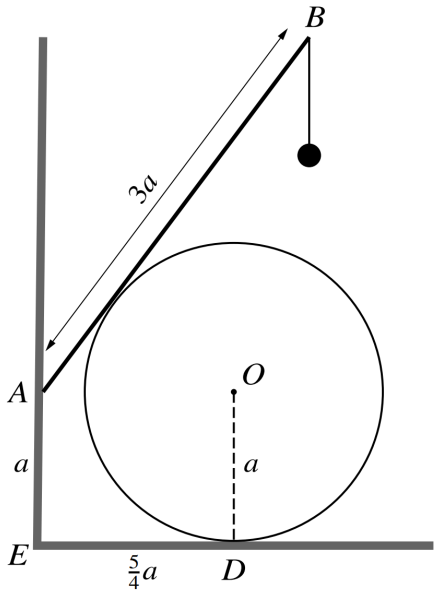
(FM) 2. Equilibrium of a rigid body11

NO. 1



A uniform smooth disc with centre O and radius a is fixed at the point D on a horizontal surface. A uniform rod of length $3a$ and weight W rests on the disc with its end A in contact with a rough vertical wall. The rod and the disc lie in a vertical plane that is perpendicular to the wall. The wall meets the horizontal surface at the point E such that $AE = a$ and $ED = \frac{5}{4}a$. A particle of weight kW is hung from the rod at B (see diagram). The coefficient of friction between the rod and the wall is $\frac{1}{8}$ and the system is in limiting equilibrium. Find the value of k. [8]

NO. 2

A uniform rod AB has length $2a$ and weight W. The end A rests on rough horizontal ground and the end B rests against a smooth vertical wall. The rod is in a vertical plane that is perpendicular to the wall. The angle between the rod and the horizontal is θ. A particle of weight $5W$ hangs from the rod at the point C, with $AC = xa$, where $0 < x < 1$.

i. By taking moments about A, show that the magnitude of the normal reaction at B is $\frac{W(5x+1)}{2\tan\theta}$. [3]

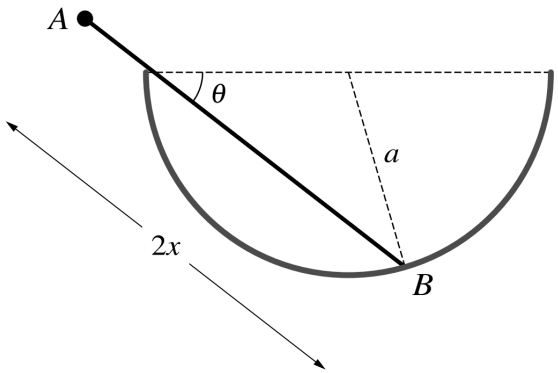
The particle of weight $5W$ is now moved a distance a up the rod, so that $AC = (x + 1)a$. This results in the magnitude of the normal reaction at B being double its previous value. The system remains in equilibrium with the rod at angle θ with the horizontal.

ii. Show that $x = \frac{4}{5}$. [3]

The coefficient of friction between the rod and the ground is $\frac{2}{3}$.

iii. Given that the rod is about to slip when the particle of weight $5W$ is in its second position, find the value of $\tan\theta$. [5]

NO. 3



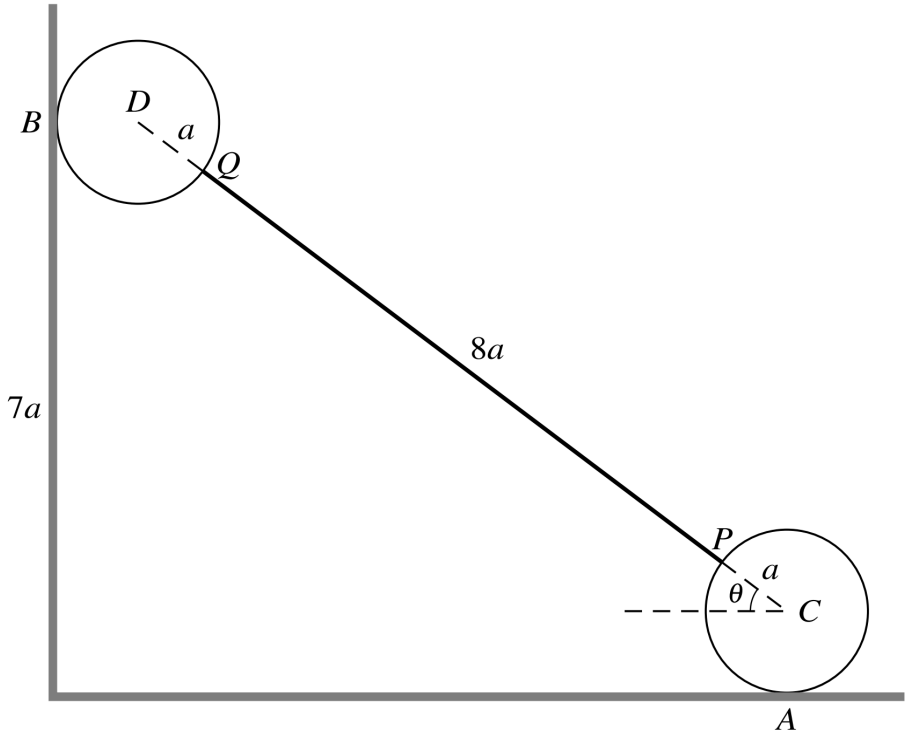
A uniform rod AB of length $2x$ and weight W rests on the smooth rim of a fixed hemispherical bowl of radius a. The end B of the rod is in contact with the rough inner surface of the bowl. The coefficient of friction between the rod and the bowl at B is $\frac{1}{3}$. A particle of weight $\frac{1}{4}W$ is attached to the end A of the rod. The end B is about to slip upwards when AB is inclined at an angle θ to the horizontal, where $\tan\theta = \frac{3}{4}$ (see diagram).

i. By resolving parallel to the rod, show that the normal component of the reaction of the bowl on the rod at B is $\frac{3}{4}W$. [5]

ii. Find, in terms of W, the reaction between the rod and the smooth rim of the bowl. [4]

iii. Find x in terms of a. [3]

NO. 4



The end P of a uniform rod PQ, of weight kW and length $8a$, is rigidly attached to a point on the surface of a uniform sphere with centre C, weight W and radius a. The end Q is rigidly attached to a point on the surface of an identical sphere with centre D. The points C, P, Q and D are in a straight line. The object consisting of the rod and two spheres rests with one sphere in contact with a rough horizontal surface, at the point A, and the other sphere in contact with a smooth vertical wall, at the point B. The angle between CD and the horizontal is θ. The point B is at a height of $7a$ above the base of the wall (see diagram). The points A, B, C, D, P and Q are all in the same vertical plane.

i. Show that $\sin\theta = \frac{3}{5}$. [1]

The object is in limiting equilibrium and the coefficient of friction at A is μ.

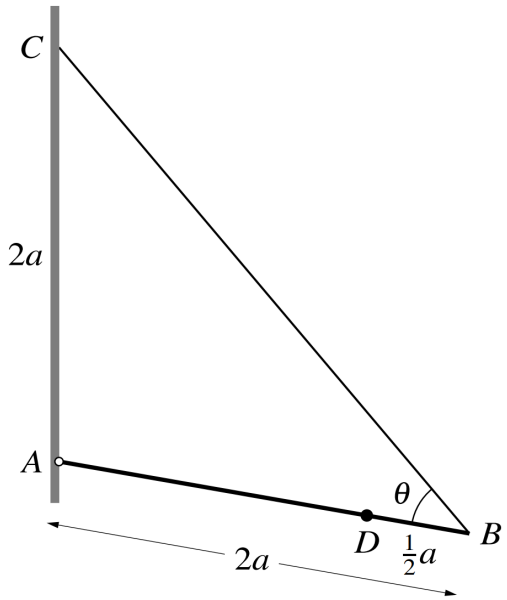
ii. Find the numerical value of μ. [7]

iii. Given that the resultant force on the object at A is $W\sqrt{(65)}$, show that $k = 5$. [3]

NO. 5

Answer only **one** of the following two alternatives.

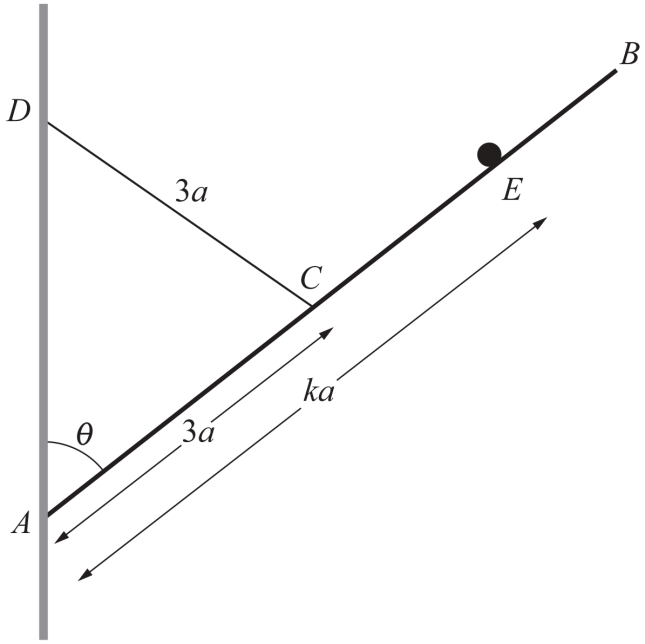
EITHER



i. The end A of a uniform rod B, of length $2a$ and weight W, is freely hinged to a vertical wall. The end B of the rod is attached to a light elastic string of natural length $\frac{3}{2}a$ and modulus of elasticity $3W$. The other end of the string is attached to the point C on the wall, where C is vertically above A and $AC = 2a$. A particle of weight $2W$ is attached to the rod at the point D, where $DB = \frac{1}{2}a$. The angle ABC is equal to θ (see diagram). Show that $\cos\theta = \frac{3}{4}$ and find the tension in the string in terms of W. [10]

ii. Find the magnitude of the reaction force at the hinge. [4]

NO. 6

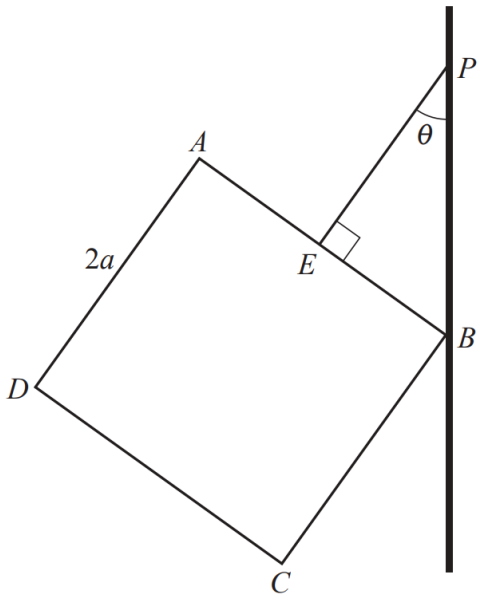


The end A of a uniform rod AB of length $6a$ and weight W is in contact with a rough vertical wall. One end of a light inextensible string of length $3a$ is attached to the midpoint C of the rod. The other end of the string is attached to a point D on the wall, vertically above A. The rod is in equilibrium when the angle between the rod and the wall is θ, where $\tan\theta = \frac{3}{2}$. A particle of weight W is attached to the point E on the rod, where the distance AE is equal to ka $(3 < k < 6)$ (see diagram). The rod and the string are in a vertical plane perpendicular to the wall. The coefficient of friction between the rod and the wall is $\frac{1}{3}$. The rod is about to slip down the wall.

a. Find the value of k. [5]

b. Find, in terms of W, the magnitude of the frictional force between the rod and the wall. [2]

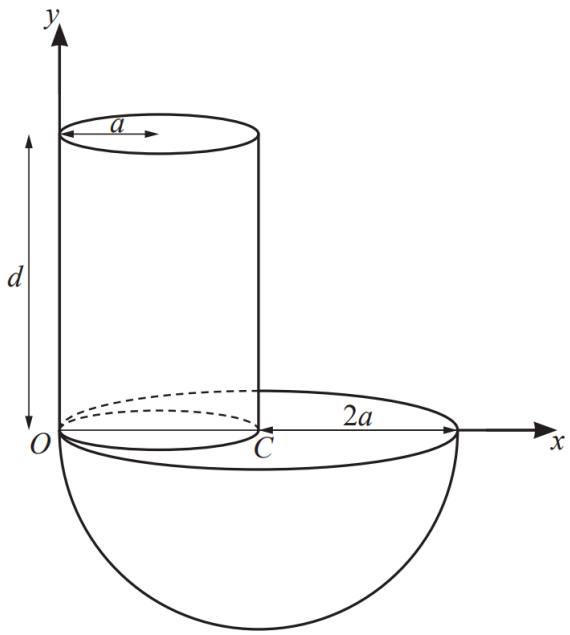
NO. 7



A uniform square lamina of side $2a$ and weight W is suspended from a light inextensible string attached to the midpoint E of the side AB. The other end of the string is attached to a fixed point P on a rough vertical wall. The vertex B of the lamina is in contact with the wall. The string EP is perpendicular to the side AB and makes an angle θ with the wall (see diagram). The string and the lamina are in a vertical plane perpendicular to the wall. The coefficient of friction between the wall and the lamina is $\frac{1}{2}$.

Given that the vertex B is about to slip up the wall, find the value of tan θ. [8]

NO. 8



An object is formed from a solid hemisphere, of radius $2a$, and a solid cylinder, of radius a and height d. The hemisphere and the cylinder are made of the same material. The cylinder is attached to the plane face of the hemisphere. The line OC forms a diameter of the base of the cylinder, where C is the centre of the plane face of the hemisphere and O is common to both circumferences (see diagram). Relative to axes through O, parallel and perpendicular to OC as shown, the centre of mass of the object is $(\overline{x},\overline{y})$.

a. Show that $\overline{x} = \dfrac{32a^2 + 3ad}{16a + 3d}$ and find an expression, in terms of a and d, for $\overline{y}$. [5]

The object is placed on a rough plane which is inclined to the horizontal at an angle θ where $\sin\theta = \frac{1}{6}$. The object is in equilibrium with CO horizontal, where CO lies in a vertical plane through a line of greatest slope.

b. Find d in terms of a. [3]